Julian Lozada

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EDUCATION

California State University
B.S. in Computer Science — Expected August 2026

PROJECTS

Sea Turtle Tracking System (Senior Capstone) — **September 2025 - Present**
California State University, Northridge
- Assisted in designing a hardware system integrating sea turtle trackers with machine learning and cloud-based data pipelines to relay real-time migration and ocean condition metrics.
- Developed a web application to visualize environmental data and turtle movement patterns for use by researchers and conservation teams, enabling monitoring of over 30 tracked turtles and directly informing tagging and intervention decisions across two active migration corridors
- Integrated IoUT communication systems to improve underwater data accuracy and reliability, achieving approximately 40% reduction in packet loss and increasing sensor read consistency to over 92% across field deployments.
- Collaborated with the University of Colombia and marine biologists to co-develop a standardized data schema and shared dashboard interface, ensuring technical solutions aligned with ecological research protocols while enabling seamless cross-instituional data sharing across both teams
- Produced a technical research paper and project documentation, contributing to the scalability and long-term sustainability of marine tracking initiatives.

TetrisAI — **February 2026**
Personal Project
- Trained a Deep Q-Network (DQN) through a reinforcement learning agent over 10,000+ episodes using PyTorch, implementing experience replay, epsilon-greedy exploration, and custom reward shaping to maximize performance
- Exported the trained neural network to ONNX format and built a live React and TypeScript browser component using ONNX Runtime Web, achieving real-time AI inference at 60fps with no backend dependency
- Delivered an end-to-end ML pipeline spanning game simulation (Python and Pygame), model training, serialization, and client-side web deployment all integrated as a live interactive route on portfolio

EcoSim — **January 2026 - Present**
California State University, Northridge
- Co-developed a real-time browser-based ecosystem simulation using React, TypeScript, Three.js, and WebGL2 simulating 1,000+ autonomous agents running entirely on the GPU
- Architected a ping-pong buffer GPU compute system using GLSL shaders, encoding agent state in floating-point textures and sustaining 60fps render performance in the browser without a backend
- Collaborated with a 4-person Agile team to deliver modular GPU simulation engine components, adhering to a shared integration contract that enabled parallel frontend and simulation development across the codebase

Portfolio Website (lozadajulian.com) — **January 2026 – February 2026**
Personal Project
- Built and deployed a production full-stack web application using React, TypeScript, Vite and Java Spring Boot, PostgreSQL, REST API, all serving as a live portfolio with multiple dynamic features
- Engineered responsive UI components with CSS animations, custom hash-based client-side routing, and a component library architecture across 10+ reusable React components
- Architected a full-stack feature pipeline connecting a React/TypeScript frontend to a Spring Boot/PostgreSQL backend, handling data flow from client requests through REST endpoints to database persistence and back

PROFESSIONAL EXPERIENCE

Volunteer Computer Science and Calculus Tutor — **September 2022 – Current**
Los Angeles Pierce College and California State University, Northridge
- Initiated tutoring support outside of official campus resources to help underclassmen with introductory programming, data structures, and algorithm coursework.
- Conducted weekly one-on-one and group sessions focused on breaking down abstract concepts into real-world examples using Java, Python, and C++, fostering a collaborative learning environment for students.
- Built custom study materials, mock exams, and debugging walkthroughs to reinforce problem-solving strategies and build confidence among peers, enabling students to approach challenging coursework effectively and independently.
- Known for adapting explanations to individual learning styles, resulting in improved student performance and engagement.

SKILLS & INTERESTS

Technical: Python, TypeScript, JavaScript, Java, SQL, HTML 5, CSS3, React, PostgreSQL, Spring Boot, WebGL2, AI Prompting